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ABOUT ME

I combine a strong statistics and machine learning foundation with hands-on experience in quantitative finance and algorithmic trading. My work includes building deep learning architectures for market prediction, applying change point detection to identify distributional shifts in financial data, and implementing real-time trading engines with risk management controls. I am particularly interested in the intersection of machine learning and finance, where I aim to design rigorous and practical models for market forecasting and analysis. Beyond my technical experience, I bring proven leadership and teamwork from serving as both a college football team captain and fraternity president.

EDUCATION

Stony Brook University

MS in Applied Mathematics and Statistics

Stony Brook, NY

August 2025 – Present

- Relevant Coursework: Machine Learning in Finance, Financial Mathematics, Stochastic Calculus, Computational Finance, Portfolio Theory, Quantitative Risk Management, Time Series Analysis

Hamilton College

BA in Mathematics and Computer Science; GPA: 3.77/4.0

Clinton, NY

August 2021 – May 2025

- Relevant Coursework: Statistical Modeling, Probability, Programming, Statistical Machine Learning, Python for Data Science, Algorithms, Financial Economics

WORK EXPERIENCE

Data Annotation

AI Expert Trainer (Part-time Contractor)

Remote, USA

May 2025 – Present

- Generate coding prompts for **Large Language Models** (LLMs) coupled with **complex codebases** for testing model correctness, instruction execution, and efficiency.
- Write solutions to coding prompts and evaluation reports for the LLM to learn from.

Oppenheimer & Co. Inc

Informational Technology Intern

NYC, NY

June 2024 – August 2024

- Developed and **optimized backend endpoints** for the financial advisor platform, enhancing system performance and user experience.
- Implemented **unit tests** to ensure the reliability and functionality of new and existing code.
- Wrote **SQL queries** to **retrieve and analyze** data from company databases, supporting backend development and ensuring data accuracy.
- Led winning intern team on an **equity research project**, **analyzing market data** to develop a **diversified portfolio** that we **presented to a board of managing directors**.

Hamilton College

Computer Science Teaching Assistant

Clinton, NY

August 2022 – May 2025

- Supported students in understanding and completing coding labs.
- Provided guidance on **programming concepts, debugging, and problem-solving**.

PROJECTS

Trading Engine

Designed and implemented a multi-asset trading engine simulator

Stony Brook University

August 2025

- Implemented a **machine learning driven risk manager** to assess regime changes, resulting in **lowered drawdowns and portfolio volatility**.
- Performed **rapid backtesting for multiple quantitative strategies**, outputting visuals and significant performance metrics with comparisons to market benchmarks.
- Integrated with **Alpaca API** for **historical asset data** to simulate trades with **bid-ask spread and slippage**.

LSTM-MLP Option Pricer

Developed a hybrid machine learning model for call option pricing

Stony Brook University

July 2025

- Built a **neural network** with **PyTorch** to improve option value forecasting by **integrating sequential and dense layers** in a custom training loop.
- Preprocessed option data using **Python (pandas)** to streamline workflow for **static and encoded features**.

CUSUM Change Point Detection Algorithm

Created an algorithm to detect volatility-based change points in financial time series.

Stony Brook University

June 2025

- Constructed a Streamlit Application to interactively illustrate volatility spike detection using **online CUSUM methodology** with **kernel-based and likelihood-based methods**.

SKILLS

Technical Skills: Machine Learning, Algorithmic Trading, Statistical Modeling, Risk Management, Data Analysis

Programming Skills: Python, R, C++, SQL, Ruby, Haskell

Software Tools: Git, Jupyter Notebook, MongoDB, Excel, $\text{\LaTeX}$